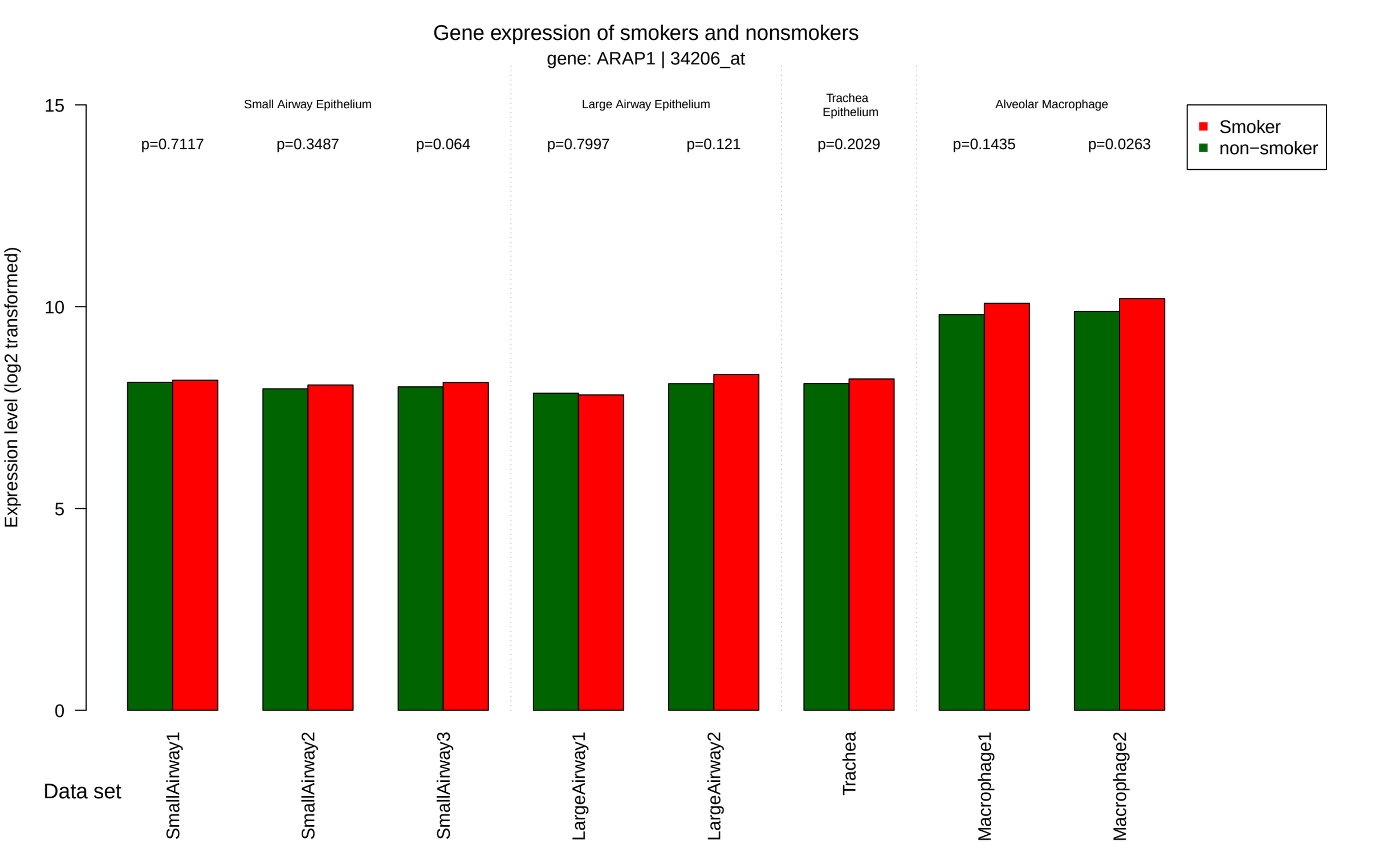
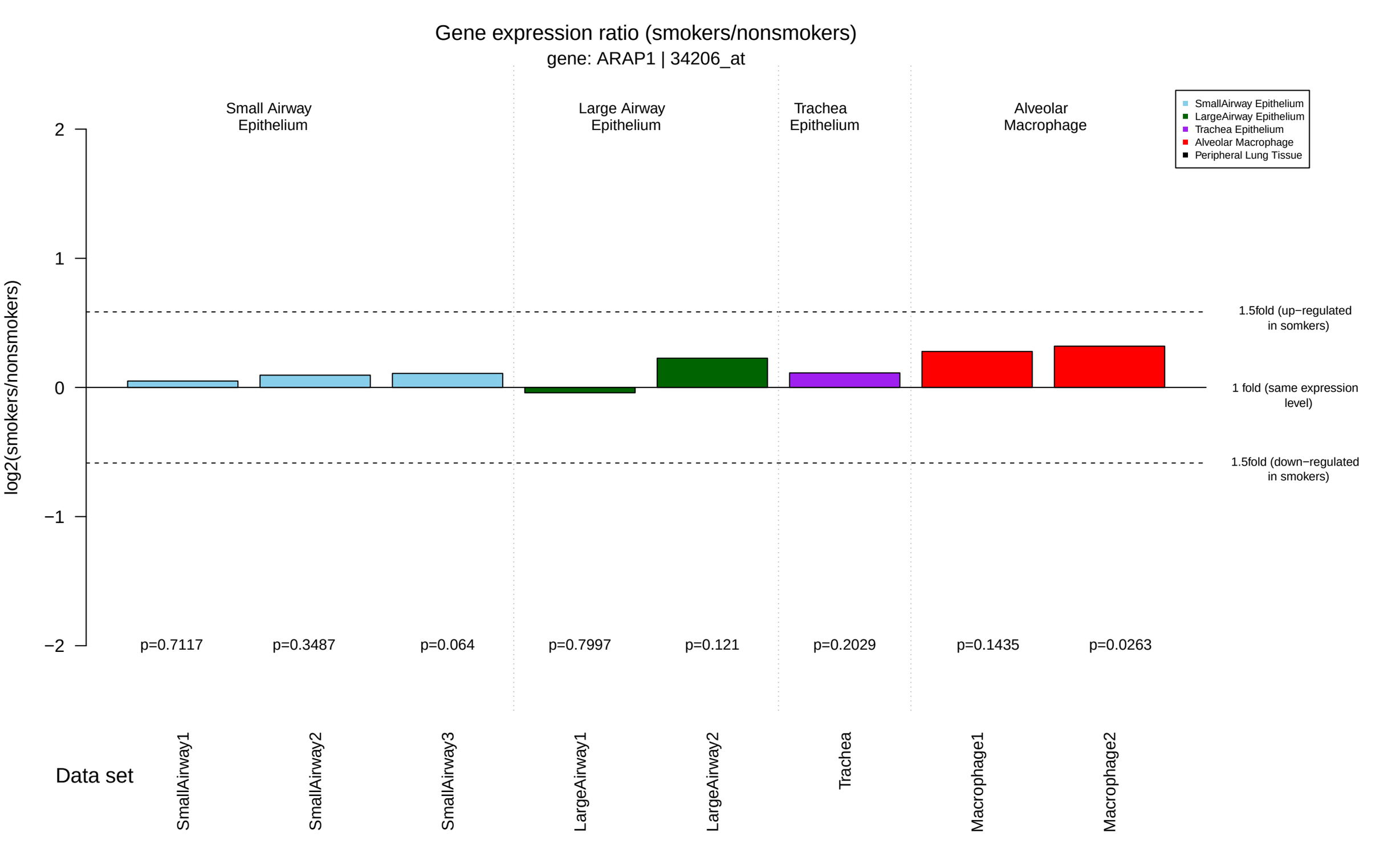
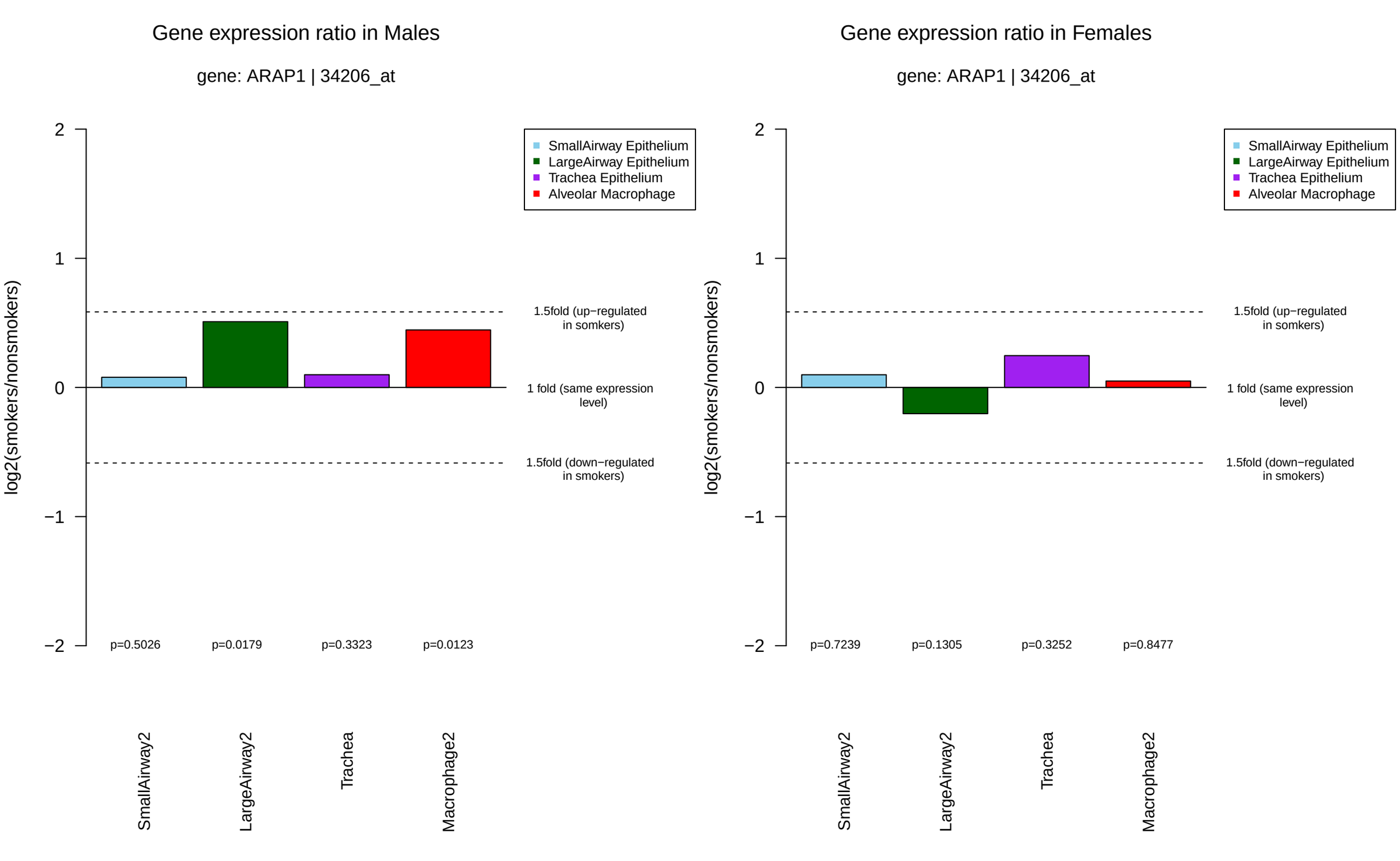
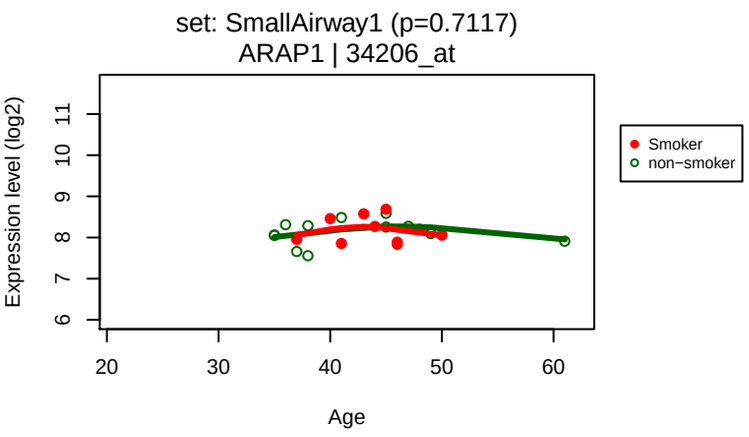
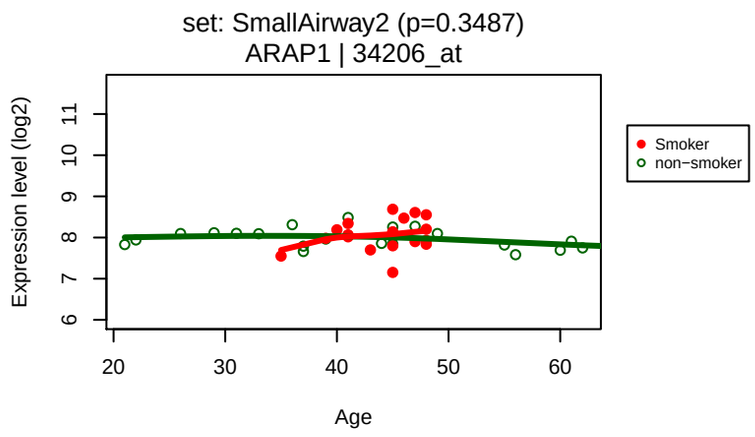
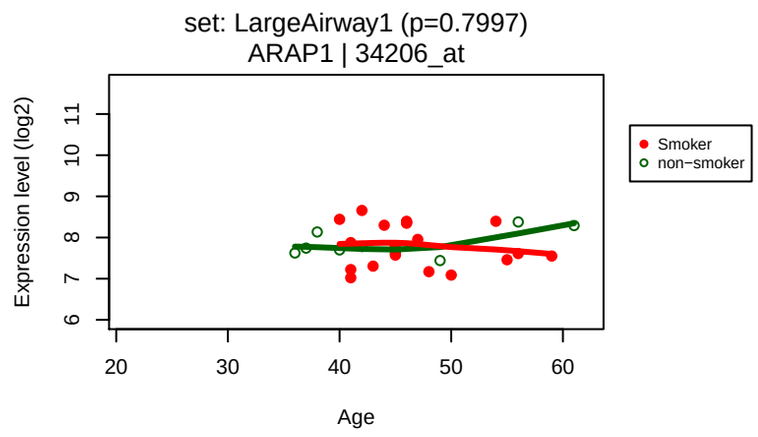

Gene expression of smokers and nonsmokers
gene: ARAP1 | 34206_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7117
p=0.3487
p=0.064
p=0.7997
p=0.121
p=0.2029
p=0.1435
p=0.0263
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

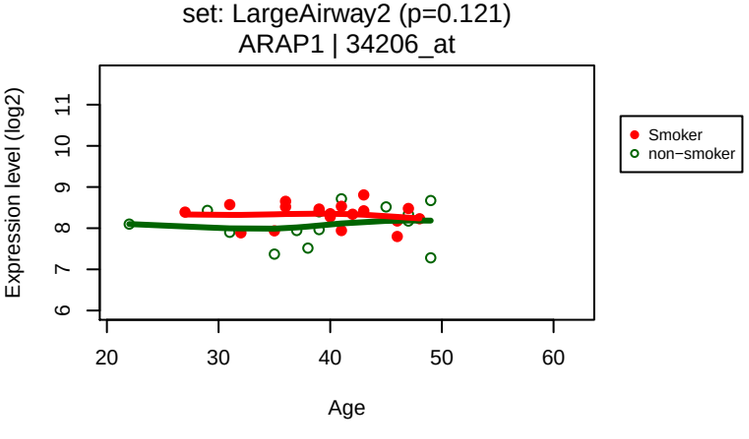

Gene expression ratio (smokers/nonsmokers)
gene: ARAP1 | 34206_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7117
p=0.3487
p=0.064
p=0.7997
p=0.121
p=0.2029
p=0.1435
p=0.0263
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

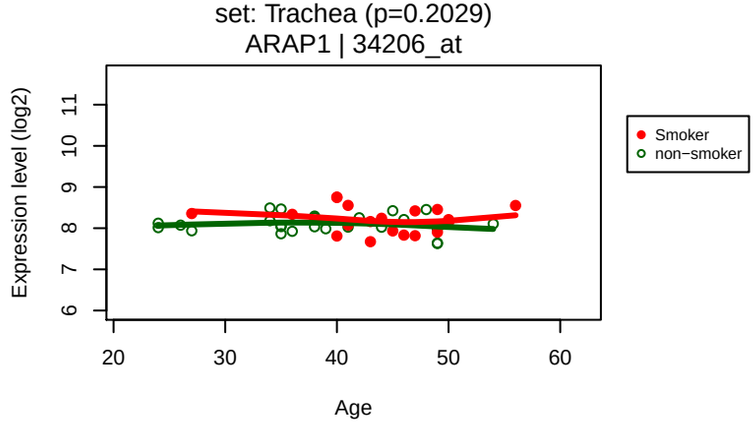

Gene expression ratio in Males
gene: ARAP1 | 34206_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5026
p=0.0179
p=0.3323
p=0.0123
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARAP1 | 34206_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7239
p=0.1305
p=0.3252
p=0.8477
SmallAirway2
LargeAirway2
Trachea
Macrophage2

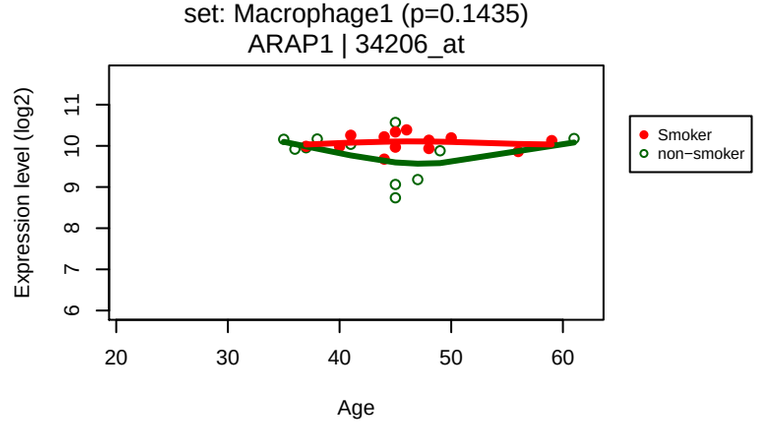

set: SmallAirway1 (p=0.7117)
ARAP1 | 34206_at
Expression level (log2)
Age
Smoker
non−smoker

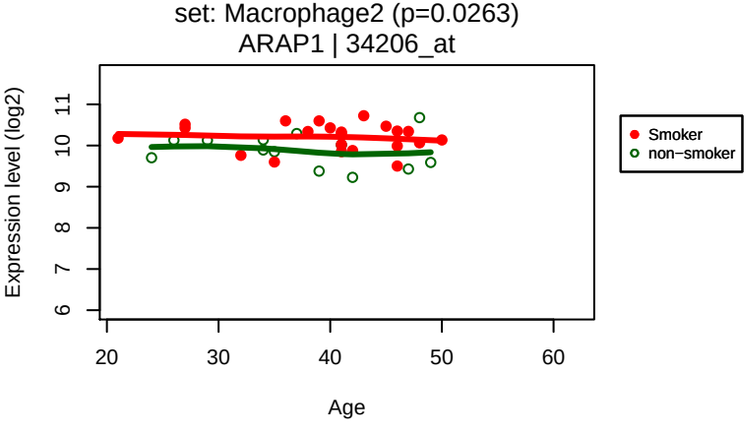

set: SmallAirway2 (p=0.3487)
ARAP1 | 34206_at
Expression level (log2)
Age
Smoker
non−smoker

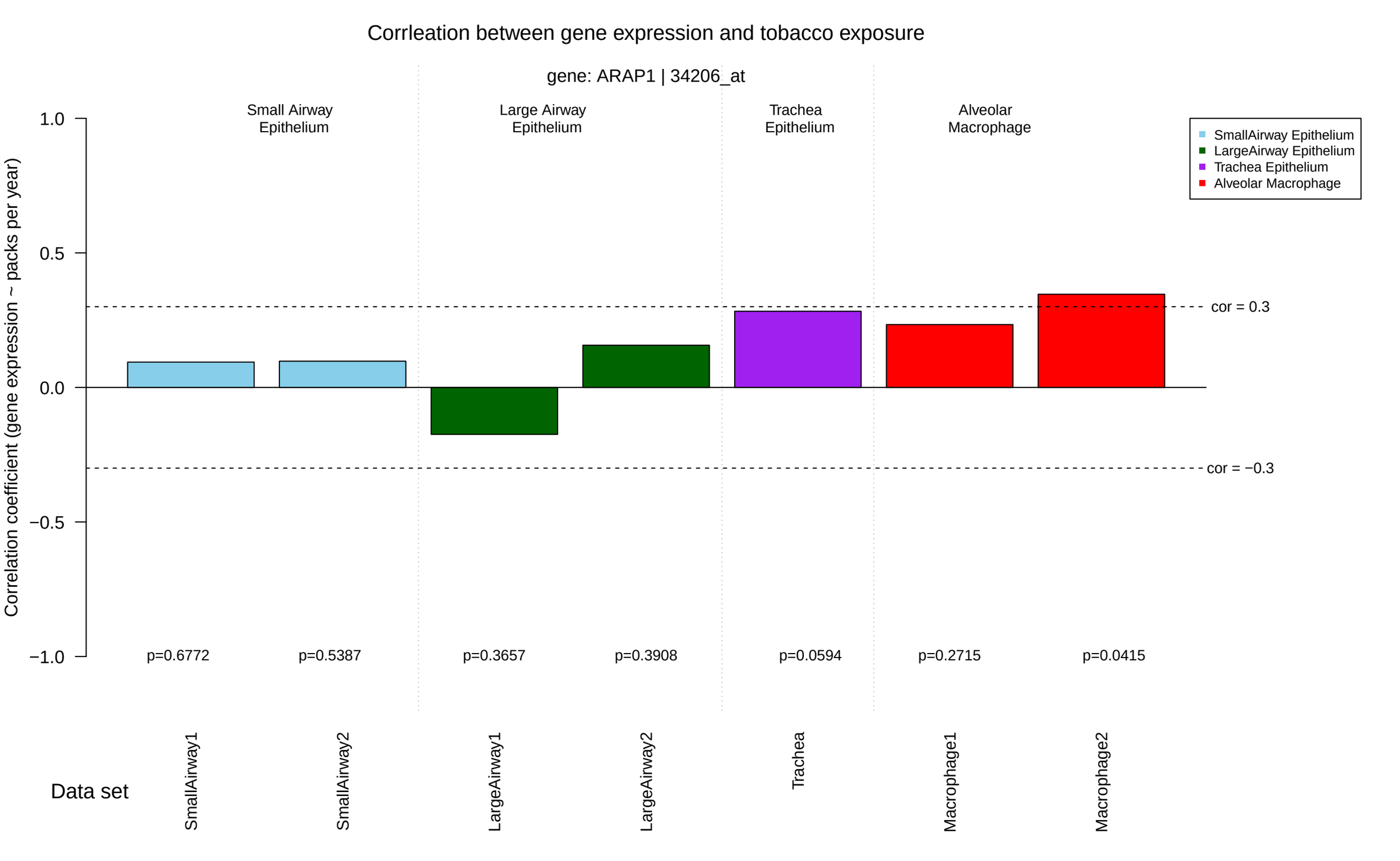

set: LargeAirway1 (p=0.7997)
ARAP1 | 34206_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.121)
ARAP1 | 34206_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2029)
ARAP1 | 34206_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1435)
ARAP1 | 34206_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0263)
ARAP1 | 34206_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARAP1 | 34206_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6772
p=0.5387
p=0.3657
p=0.3908
p=0.0594
p=0.2715
p=0.0415
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2